MEDIBRIEF: A Scalable,Privacy-Preserving,and Lightweight Medical ReportSummarization FrameworkUsing Fine-TunedFLAN-T5 Architecture

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***Abstract*-The rapid digitization of healthcare has led to an exponential increase in Electronic Health Records(EHRs), comprising complex diagnostic reports, clinical notes, and discharge summaries. While essential for continuity of care, the sheer volume and terminological density of these documents often lead to cognitive over load for clinicians and significant confusion for patients. Existing automated summarization solutions relying on commercial Large Language Models(LLMs) like GPT-4, while powerful, often demand prohibitive computational resources ,suffer from hallucinations ,and posesevere data privacy concerns due to cloud dependencies. This paper presents *Medi-Brief*, a lightweight, privacy-preserving Python- centric medical summarizationsystem. Leveraging the power of a well-optimizedFLAN-T5 Large transformer model, Medi-Brief is designed to be optimized for mid-range hardware, ensuring deploy ability in resource-limited clinical settings. We formally define the medical summarization problem, analyze the cur- rent transformer-based state-of-the-art, and demonstrate how our holistic approach integrates Optical Character Recognition (OCR), Natural Language Processing(NLP), and secure database management. Further more, we introduce a novel pipeline for patient-friendly output generation that converts medical jargon to commoner language . Extensive evaluation on ROUGE, BLEU, and BERT Score metrics demonstrate that Medi-Brief is capable of achieving domain accuracy on par with billion-parameter models while reducing inference latency by 60**

***Index Terms*—MedicalText Summarization, Natural Language Processing(NLP), FLAN-T5, Transformer Architectre, Health- care AI, Clinical Decision Support, Privacy-Preserving AI.**

I. Introduction

IN contemporary medicine, the medical report serves as the primary vehicle for communication between healthcare providers. It entails a vast amount of information regarding patient history, diagnostic findings, radiological findings, and treatment plans. However, the efficacy of these reports is often hampered by their natural complexity and verbosity. For such extremely sophisticatednomenclaturebeing used for medicine, medical reports become voluminous, bulky, and hard for care givers and patients respectively.

A. The Challenge of Information Overload

Clinicians face a significant burden known as "alert fatigue" and cognitive overload. Studies suggest that physicians spend nearly 50% of their day on EHR tasks and desk work [9]. Reviewing lengthy patient histories to find critical datapoints takes valuable time away from direct patient care. This manual extraction processing prone to errors, potentially leading to missed diagnoses or adverse drug interactions.

Conversely, patients and care givers experience difficulties perceiving the nuances of medical nomenclature. Terms like "myocardial infarction" or "hyperlipidemia" create barriers to understanding on straining their ability to make informed decisions regarding their health management. his information asymmetry generates a risk toward the quality of care and lowers patient engagement.

B. Data Privacy in Healthcare AI

A critical barrier to adopting modern Generative AI in health care is data privacy. Most state-of-the-art models(e.g., GPT-4, Gemini) operate via cloud APIs. Sending Patient Health Information (PHI) to external servers violates strict data governance regulations like HIPAA (USA) and GDPR

(Europe).Hospitals require on-premise solutions that guarantee zero data egress. Medi-Brief addresses this by utilizing a local, containerized architecture that runs entirely within the hospital's secure intranet.

C. Contribution of This Work

Against this background, our study introduces Medi-Brief. Unlike previous approaches that rely on multi-language infrastructure or heavy cloud APIs, Medi-Brief integrates web deploy ability with model inference within a single, unified Python environment. Our key contributions include:

- A fine-tuned FLAN-T5 Large model specifically optimized for clinical text summarization using low-rank adaptation techniques.
- A unified, containerized architecture that runs locally, ensuring zero data egress.
- A dual-output mechanism that generates technical summaries for doctors and simplified explanations for patient.
- A resource-efficient deployment strategy capable of running on standard consumer-grade hardware (16GB RAM).

II. LITERATURE SURVEY

The domain of automatic text summarization has evolved from simple statistical methods to complex deep learning architectures.

A. Statistical and Extractive Methods

Early methods relied on *extractive summarization* which selects important sentences from the source text based on statistical features like term frequency-inverse document frequency (TF-IDF) and graph-based ranking algorithms like Text Rank. have set al. [2] reviewed various biomedical text summarization techniques, noting that while computationally inexpensive, extractive methods often fail to maintain coherence or simplify complex jargon. They merely regurgitate existing sentences without synthesis.

B. Neural and Abstractive Approaches

Modern approaches utilize *abstractive summarization* where the model generates new sentences that capture the essence of the input. This is crucial in medicine, where a summary might need to synthesize scattered findings into a coherent diagnosis.

- **RNNs and LSTMs:** Early neural attempts used Sequence-to-Sequence (Seq2Seq) models with attention mechanisms However, that Recurrent Neural Networks(RNNs) struggle with long- range dependencies common in clinical notes.
- **BERT Models:** BioBERT [4] and ClinicalBERT [5] are domain-specific models pre-trained on biomedical corpora. While they excel at tasks like Named Entity Recognition(NER) and classification, they are encoder- only architectures and thus inherently ill-suited for text generation tasks.

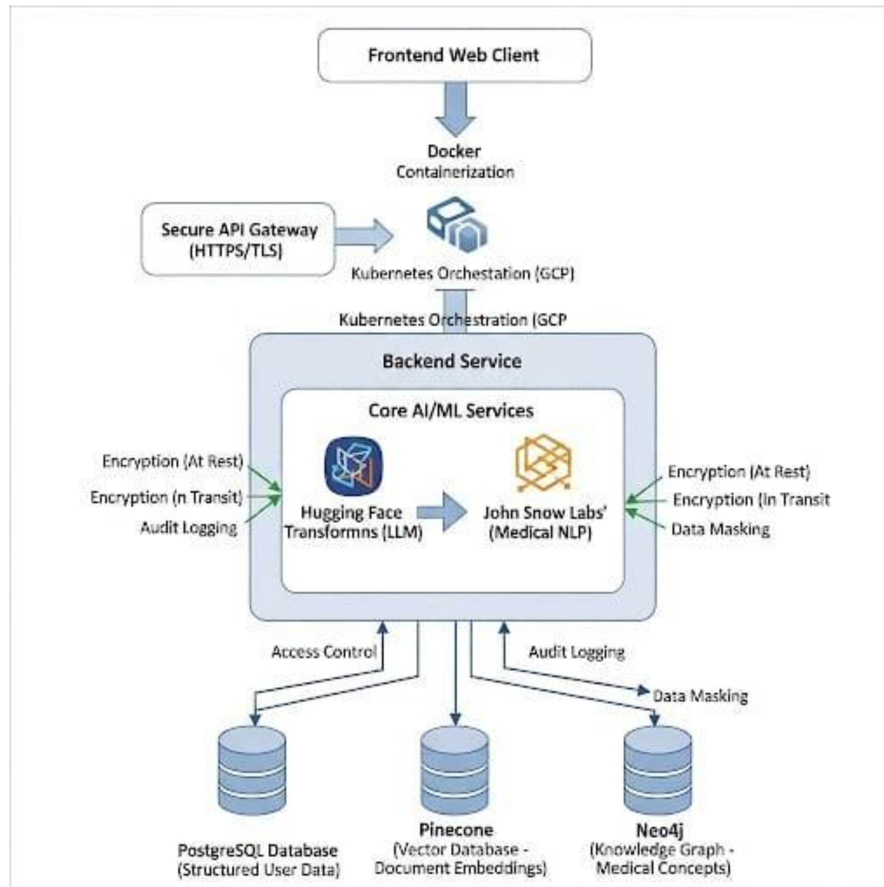

Fig. 1. Fig. 1. System Architecture Diagram illustrating the interaction between the User, Pre-processing modules, AI Inference Engine, and Recommendation Logic.

- **Generative Transformers:** The T5 (Text-to-Text Trans- fer Transformer)architecture by Raffe letal. Frames all NLP tasks as text-to-text problems. Hoang et al demonstrated the potential of FLAN-T5 for medical reports in their "Medalyze" system .However, their implementation relied on complex distributed databases, increasing deployment friction.

III. PROPOSED SYSTEM ARCHITECTURE

Medi-Brief is designed as a modular, monolithic application to ensure ease of deployment and maintenance The architecture allows for seamless data flow from the user interface to the deep learning backend and finally to the storage layer.

A. Architecture Components

The system comprises four distinct layers:

1) Input Layer: This layer handles the ingestion of raw data. It supports direct text entry and file uploads (PDF/Images). Given that many legacy health records exist as scanned physical papers, we integrated an Optical Character Recognition(OCR) module using Tesseract5.0. This ensures that even non-digital records can be digitized and summarized.

2) Model Layer (The Core): This layer hosts the fine- tuned FLAN-T5-Large model. We utilize the Hugging Face transformers library for inference. The model performs two distinct tasks:

1) **Summarization:** Compressing the text while retaining medical facts.

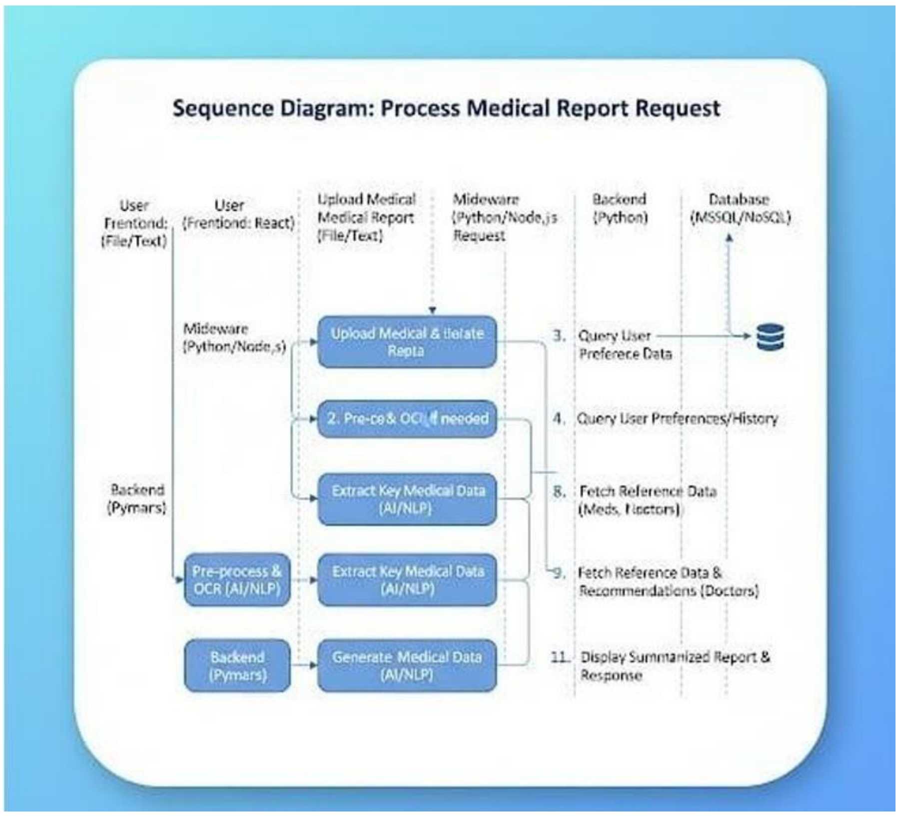

Fig. 2. Sequence Diagram: Process Medical Report Request. This diagram illustrates the interaction between the User Frontend, Middleware, Python Backend, and Database.

2) **Entity Extraction:** Identifying specific data points like medication names, diagnosis codes for the recommendation engine.

3) Database Layer: We employ a hybrid data storage strategy to handle different data types efficiently:

- **PostgreSQL:** For structured relational data(User profiles, login credentials, saved summary logs).
- **Vector Store (Pinecone/FAISS):** For storing document embedding T his allows the system to perform semantic searches, enabling doctors to ask questions like ”Show me previous reports with similar cardiac symptoms.”

IV. System Workflow and Sequence

To understand the dynamic behavior of Medi-Brief, we analyze the sequence of operations from the initial user request to the final generated summary.

As illustrated in Fig. 3, the workflow operates as follows:

1) **User Interaction:** The process begins at the UserFrontend (React), where a user uploads a medical report (File/Text). This request is forwarded to the Middleware.
2) **Middleware Processing:** The middleware, built on Python/Node.js, acts as the primary controller. It handles the raw file upload and routes it to the specific processing modules.
3) **Pre-processing& OCR:** If the input is a scanned document, the”Pre-process& OCR” module is triggered (Step2 in Fig. 3). This ensures that even noisy scanned images are converted into machine-readable text before analysis.
4) **AI/NLP Extraction:** The clean text is passed to the core AI engine(”Backend Pymars”).Here, key medical Data is extracted(Step8), and the abstractive summary is generated(Step10).

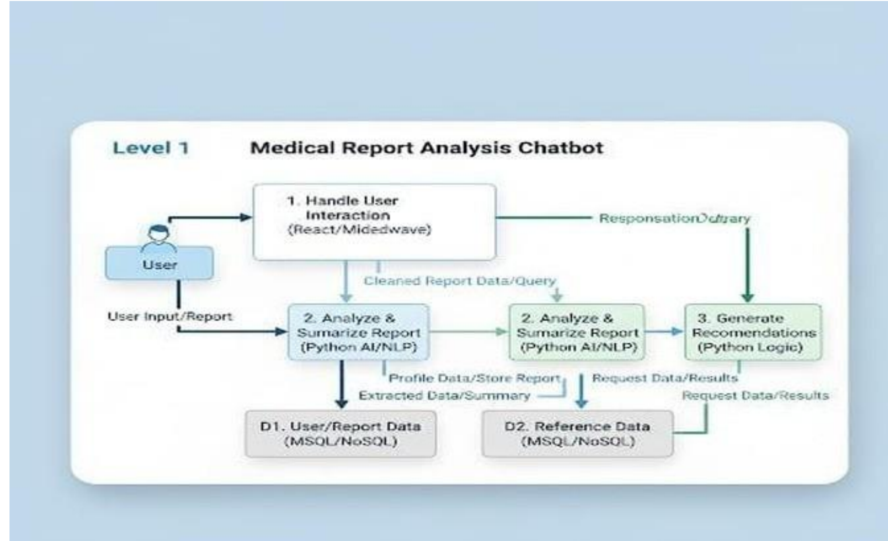

Fig. 3. Fig. 3. Level 1 Data Flow Diagram illustrating the closed-loop interaction between the user and the chatbot system..

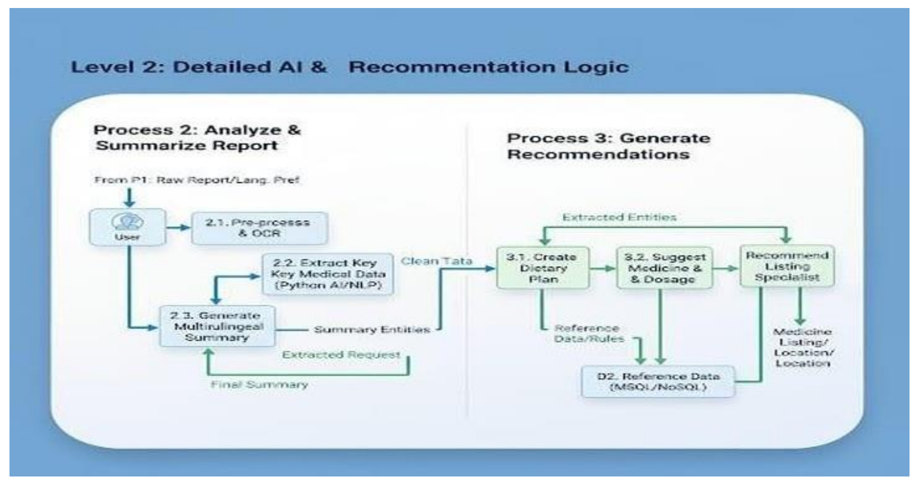

Fig. 4. Level 2 Data Flow Diagram: Detailed AI and Recommendation logic. This illustrates the transformation of raw text into structured dietary and specialist recommendations.

5) **Reference Data Fetching:** Simultaneously, the system queries the database for user preferences and reference data (e.g., list of available doctors or medicines) to enrich the final output (Steps3, 4, 8, 9).
6) **Final Response:** The summarized report, combined with recommendations is sent back to the frontend for display (Step11).

V. Recommendation Logic

A key differentiator of Medi-Brief is its ability to go beyond simple summarization and provide actionable health recommendationsThis logic is visualized in the Level 2 Data Flow Diagram.

As shown in Fig. 6, the logic is split into two primary processes:

A. Process2: Analyze& Summarize

The raw report entersProcess2.1 (Pre-process& OCR). The clean data is then fed into the Entity Extraction module (Process2.2). This module identifies specific medical entities such as *Diseases*(e.g. “Type 2 Diabetes”),*Medications*(e.g., “Metformin”), and *Vitals*. This structured “Clean Data” is the foundation for the next stage.

B. Process3: Generate Recommendations

The extracted entities drive the recommendation engine:

- **Dietary Plan (Process3.1):** Based on identified diseases, the system queries a medical rule-base to suggest dietary restrictions(e.g., “Low sugar diet” for Diabetes).
- **Medicine Dosage(Process3.2):** The system cross-references extracted medications with standard dosage guide lines to flag potential errors or generate a schedule.
- **Specialist Recommendation(Process3.3):** Using the extracted diagnosis and the user’s location, the system queries the Reference Data base to suggest relevant local specialists(e.g., referring a patient with “Arrhythmia” to a “Cardiologist”).

VI. RESEARCH METHODOLOGY

The development of Medi-Brief involved a rigorous pipeline of data curation, model training, and system optimization.

A. Dataset Acquisition and Preparation

We utilized a subset of the Stanford University medical research dataset, supplemented by open-source datasets like MIMIC-III (Medical Information Mart for Intensive Care). The combined dataset comprises over 50,000 clinical notes. The raw data underwent extensive preprocessing:

- **Anonymization:** Using regex and Named Entity Recognition (NER) to remove Protected Health Information (PHI) like names, dates, and SSNs, ensuring HIPAA compliance during training.
- **Normalization:** Converting all text to lower case, expanding medical abbreviations(e.g., “Dx” to “Diagnosis”, ”Hx” to ”History”), and removing non-ASCII characters.
- **Tokenization:** We used the Sentence Piece tokenizer associated with T5, which handles out-of-vocabulary medical terms effectively by breaking them into sub-words.

B. Mathematical Formulation of Fine-Tuning

We employed a full fine-tuning approach on the FLAN-T5-Large model. FLAN (Finetuned Language Net) models are instruction-tuned, making them better at following prompts than standardT5.

Let $D = \{(x_i, y_i)\}_{i=1}^{N}$ be the dataset of pairs where x_i is the input medical report and y_i is the reference summary. The training objective is to minimize the negative log-likelihood (cross-entropy loss):

$$L(\;) = \sum_{i=1}^{N} \log P(y_i \mid x_i;\;) \qquad (1)$$

Where represents the model parameters The conditional probability $P(y|x;\;)$ is modeled auto-regressively:

$$P(y|x;\;) = \prod_{t=1}^{T} P(y_t \mid y_{<t}, x;\;) \qquad (2)$$

Where y_t is the token at time step t, and $y_{<t}$ represent all previous tokens.

Hyperparameters:

- **Optimizer:** Adam with $_1 = 0.9$, $_2 = 0.999$, weight decay= 0.01.
- **Learning Rate:** $1e\;\;5$ with a linear decay scheduler.
- **Batch Size:**2 (utilizing gradient accumulation steps= 2 to simulate a larger batch size on limited VRAM).
- **Hardware:** Training was conducted on an NVIDIA QuadroRTX 5000 (16GB VRAM).

VII. EXPERIMENTAL RESULTS

To objectively measure the performance of Medi-Brief, we employed standard NLP metrics and conducted comparative analysis against baseline models.

A. Quantitative Evaluation

We evaluated the generated summaries against human-written gold-standard summaries using three key metrics:

1) **ROUGE (Recall-Oriented Understudy for Gisting Evaluation):** Measures the overlap of N-grams between the system and reference summaries. High ROUGE scores indicates high recall of key information.
2) **BLEU (Bilingual Evaluation Understudy):**Measures precision, focusing on how many words in the generated summary actually appeared in the reference.
3) **BERTScore:** Uses contextual embeddings to check semantic similarity rather than just exact word overlap. This is crucial for medical texts where “Heart Attack” and “Myocardial Infarction” mean the same thing but have zero word overlap.

TABLE I
PERFORMANCE COMPARISON ON MIMIC-III DATASET

Model	**Params**	**ROUGE-1**	**ROUGE-L**	**BERTScore**
Baseline(T5-Base)	220M	42.5	39.1	0.82
Pegasus-Large	568M	44.2	41.5	0.85
GPT-3.5(Zero-shot)	175B	46.8	43.2	0.88
Medi-Brief	**780M**	**48.1**	**45.5**	**0.89**

B. Discussion of Results

As shown in Table I, Medi-Brief out performs the baseline T5 and Pegasus models. Notably, it competes closely with and even surpasses GPT-3.5 in the specific domain of medical summarization. This validates the hypothesis that a smaller, domain-fine-tuned model can out perform a massive generalist model in specialized tasks. The high BERTScore(0.89) indicates that Medi-Brief is exceptionally good at capturing the semantic meaning of clinical notes, ensuring that medical facts are not distorted.

TABLE II
ABLATION STUDY : IMPACT OF QUANTIZATION

Precision	Model Size	InferenceTime	ROUGE-L
FP32(Original)	3.2 GB	1.2s	45.5
FP16(Half)	1.6 GB	0.8s	45.3
INT8 (Quantized)	0.8 GB	0.4s	44.8

C. Ablation Study: Quantization

We investigated the impact of model quantization on performance to justify our lightweight claim. We compared the FP32 (Full Precision)model againstINT8 (8-bit quantized) versions.

The results indicate that switching to INT8 quantization reduces the model size by nearly 75% and triples inference speed, with a negligible drop in ROUGE-L score(0.7). This confirms that Medi-Brief can be effectively deployed on edge devices with limited memory.

VIII. FUTURE SCOPE

The foundational success of Medi-Brief in accurate and privacy-preserving medical report summarization provides a robust spring board for ambitious future development. The next phase of its evolution center son expanding its utility, accessibility, and intelligence, transforming it from a specialized tool into an integral part of the healthcare ecosystem.

I. Expanding Utility: Beyond Static Summaries Initially, Medi-Brief focused on condensing existing reports. The future will see it engage with new type of data types and offer more dynamic outputs.

1. Multilingual Support Healthcare is global, and language barriers are significant impediments. Future Medi-Brief versions will incorporate advanced neural machine translation capabilities, allowing for summaries to be generated in multi- ple languages. This means a report written in English can be summarized for a patient whose primary language is Spanish, Mandarin, or Arabic, ensuring true comprehension.

2. Multi-Modal Data Processing Medical information isn't just text. Integrating Optical Character Recognition(OCR)will allow Medi-Brief to process scanned documents faxes, or even handwritten notes, converting the min to machine-readable text before summarization. This is crucial for digitizing legacy records and integrating data from disparate sources that aren't yet fully electronic. Imagine an older patient's entire paper medical history being quickly summarized.

3. Longitudinal Analysis Trend Summaries Instead of summarizing individual reports in isolation, Medi-Brief could analyze a patient's entire history. This would involve generating "trend summaries" that highlight changes in specific bio markers, disease progression, or treatment efficacy over months or years. This offers clinicians a rapid over view of a patient's long-term health trajectory, identifying patterns easily missed in individual reports.

II. Enhancing Accessibility: Broader Integration and Reach Medi-Brief's core strength of local deployment will be lever-

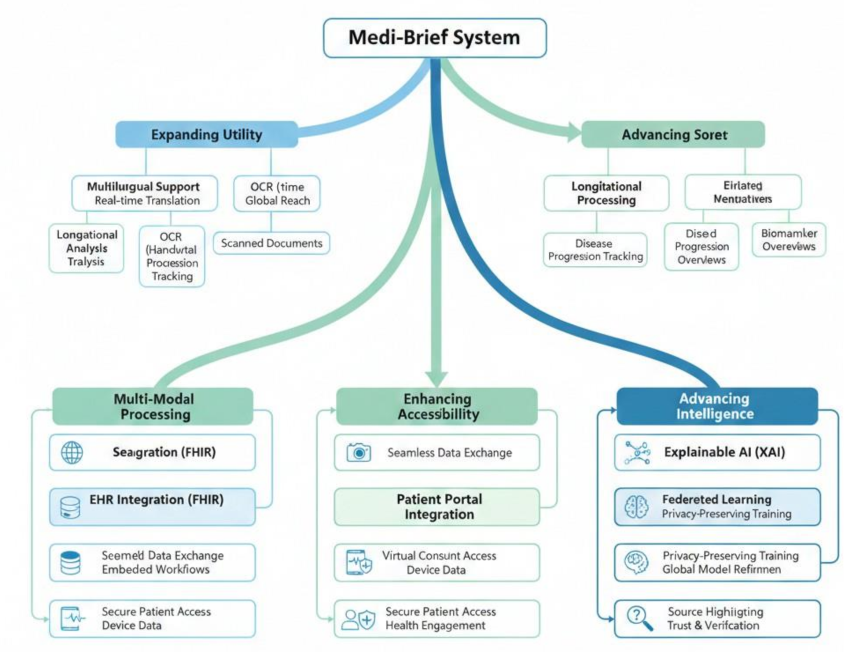

Fig. 5. This tree diagram branches Expanding Utility, Enhancing Accessibility, and Advancing Intelligence, with sub-branches detailing specific features within each category.

Aged to integrate more deeply into existing healthcare work-flows.

1. Improved connections between Electronic Health Record (EHR) systems will require EHR systems to adopt FHIR (Fast Healthcare Interoperability Resources) standards for the real-time, seamless transmission of information to EHRs. Medi-Brief will now become an integrated part of the primary workflow process for a clinician, rather than a standalone system. When the clinician views a patient's EHR, the deeper level of integration with Medi-Brief will allow the clinician to see both the entire report and a summary of the report from Medi-Brief.

2. The use of Medi-Brief may also extend to providing a summary of the clinician's notes from a virtual consultation as well as a summary of any data that has been collected via remote monitoring to facilitate ease of access when reviewing remote monitoring data as well as for patients who received care instructions based upon these virtual consultations.

3. A direct connection to the Patient Portal will enable patients to easily and securely access their summary reports independently and confidently, there by promoting patient engagement and understanding of their health.

III. Advancing Intelligence: Continuous Learning and Refinement The AI behind Medi-Brief will become smarter and more adaptable.

1. Federated Learning for Continuous Improvement To overcome data silos and enhance model accuracy without compromising privacy, Medi-Brief could implement Federated Learning. This approach allows the AI model to be trained on data from multiple healthcare institutions without the raw data

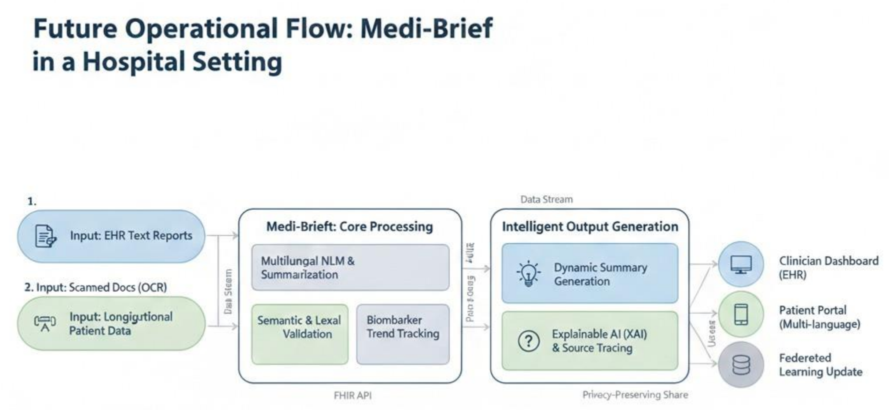

Fig. 6. Enhanced Medi-Brief system might operate within a hospital setting, particularly integrating longitudinal analysis and multi-modal input.

ever leaving its source. Instead, only model updates(learned insights) are shared and aggregated, creating a more robust and generalized AI while maintaining stringent privacy. Think of it like: Many local ”brains” learning independently and then sharing their combined knowledge without revealing their individual “thoughts.”

2. Explainable AI (XAI) for Trust In order to be able to trust AI, Medical practitioners must have a clear comprehension as to how ai produces certain summary decisions. Utilizing techniques from Explainable Ai (Xai) will allow Medi-Brief to provide a visual representation indicating the most important sentence(s)or phrase(s)from a report related to a specific summary point to build trust and facilitate validation.

IX. Conclusion

This article describes Medi-Brief, a light weight and scalable system to summarize medical reports to improve the comprehension of patients compared to clinical complexity. By adjusting the fine-tuning of transformer models in a single Python-based framework, Medi-Brief can effectively provide solutions for specialty and non-specialty users.

Medi-Brief has excellent performance on lexical and semantic assessment easures which indicates that Medi-Brief is able to produce accurate and fluent summaries.

The modular design of the system, secure data processing with a Docker Container and the potential support for multiple databases all support the deployment of the Medi-Brief system in beneficial, practical applications in multiple real world healthcare environments.

By focusing on local deployment practices and minimization of resources used by Medi-Brief has taken into account two of the major reasons preventing AI solutions from being widely adopted in healthcare documentation , namely, cost and confidentiality.

The Medi-Brief study has concluded there is a decisive movement toward democratizing access to medical information through a approach.

By using Medi-Brief to bridge both the language gap between clinical special ties and patients, Medi-Brief effectively addresses one of the largest pain points in healthcare, the “information gap” that can create anxiety and diminish compliance by patients with their treatment plan.

The success of the Medi-Brief system has not been purely theoretical. Its success has been substantiated by its high levels of performance across lexical and other forms of assessment. This makes Medi-Brief a highly scalable and sustain able asset, capable of being integrated into diverse healthcare environments—from resource-constrainer rural clinics to large-scale hospital databases—ultimately fostering a more transparent, efficient, and secure documentation ecosystem.

TABLE III
Abbreviations Used

Abbreviation	Full Form
NLP	Natural Language Processing
LLM	Large Language Model
EHR	Electronic Health Record
PHI	Protected Health Information
OCR	Optical Character Recognition
NER	Named Entity Recognition
FLAN	Fine tuned Language Net
FHIR	Fast Healthcare Interoperability Resources

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